

# GATE DA - Weekly Mock Test

Covering Week 1 (Days 1–6): Probability · Machine Learning · DBMS · Quantitative Aptitude

**Pattern:** This paper mirrors the real GATE DA exam structure exactly — 65 questions, 100 marks, split into a General Aptitude section and a core Data Science & AI section.

**Total Questions:** 65 | **Total Marks:** 100 | **Suggested Duration:** 180 minutes

**Section 1 - General Aptitude (GA):** Q1–Q10 ( $5 \times 1$  mark +  $5 \times 2$  marks = 15 marks) — drawn from Percentages, Ratio & Proportion, Averages, Profit/Loss/Discount, Time & Work, and Time/Speed/Distance.

**Section 2 - Data Science & AI Core:** Q11–Q65 (55 questions, 85 marks) — covering all 18 core topics taught this week, grouped below into three tracks (Probability & Statistics, Machine Learning, and DBMS) purely for your own revision clarity. In the real exam these would be shuffled together.

**Marking scheme (identical to the real GATE DA exam):**

- **MCQ** (1 correct option): +1 or +2 marks for a correct answer;  $-1/3$  mark for a wrong answer on a 1-mark MCQ,  $-2/3$  mark on a 2-mark MCQ. No marks for unattempted questions.
- **MSQ** (one or more correct options): no negative marking; typically requires ALL correct options selected (and no incorrect ones) for full credit.
- **NAT** (numeric answer, typed in): no negative marking.

**How to use this paper:** Attempt Section 1 and Section 2 under timed conditions first, without looking at the answer key. Mark your answers separately, then check them against the Answer Key and full Solutions at the end of this document, and compute your score using the marking scheme above.

**A note on sourcing:** Wherever a question below is a genuine, previously-documented GATE question, it is marked [VERIFIED PYQ] with its exact year and paper in the Solutions section at the end (never inside the question paper itself, to avoid giving away familiarity cues). Two topics in this week's plan — *Conditional Probability & Bayes' Theorem* and *EER + ER-to-Relational Mapping* — have no verified GATE DA PYQ available at this level of granularity, so all questions for those two topics are freshly written in GATE style. Every other question is either a verified PYQ or a GATE-style question, exactly as labelled in the Solutions section.

## Section 1: General Aptitude (Q1 - Q10)

*Percentages · [NAT · 1 mark]*

Q1. In an election between two candidates, one candidate got 60% of the total valid votes. If 8000 valid votes were cast in total, find the number of votes the winning candidate received.

*Percentages · [MCQ · 2 marks]*

Q2. A number is increased by 20% and then decreased by 20%. Find the net percentage change.

- (A) 0%
- (B) -4%
- (C) +4%
- (D) -20%

*Ratio & Proportion · [NAT · 1 mark]*

Q3. In a class, the ratio of girls to boys is 3:5. If there are 24 girls, how many boys are there?

*Ratio & Proportion · [MCQ · 2 marks]*

Q4.  $p:q = 1:2$ ,  $q:r = 4:3$ ,  $r:s = 4:5$ . If  $u$  is 50% more than  $s$ , find  $p:u$ .

- (A) 16:45
- (B) 8:15
- (C) 45:16
- (D) 15:8

*Averages · [NAT · 1 mark]*

Q5. Find the average of the first 10 natural numbers.

*Averages · [MCQ · 2 marks]*

Q6. The average weight of 8 people increases by 2.5 kg when a new person replaces one of them who weighed 60 kg. Find the weight of the new person.

- (A) 80 kg
- (B) 70 kg
- (C) 65 kg
- (D) 75 kg

*Profit, Loss & Discount · [NAT · 1 mark]*

Q7. A shopkeeper buys an item for Rs. 400 and sells it for Rs. 500. Find the profit percentage.

*Profit, Loss & Discount · [MCQ · 2 marks]*

Q8. A shopkeeper marks an item 40% above cost price and then gives a discount of 10% on the marked price. Find the overall profit percentage.

- (A) 26%
- (B) 30%
- (C) 25%
- (D) 24%

*Time & Work · [NAT · 1 mark]*

Q9. A can complete a work in 10 days, and B can complete the same work in 15 days. If they work together, how many days will they take to complete the work?

*Time, Speed & Distance · [MCQ · 2 marks]*

Q10. Two trains start from the same point and travel in OPPOSITE directions at 60 km/h and 40 km/h respectively. After how many hours will they be 300 km apart?

- (A) 3
- (B) 2
- (C) 4
- (D) 5

## Section 2: Data Science & AI Core (Q11 - Q65)

### Track A — Probability & Statistics (Q11-Q29)

*Counting & Basic Probability · [MCQ · 2 marks]*

Q11.  $n$  couples are invited to a party with the condition that every husband should be accompanied by his wife; however, a wife need not be accompanied by her husband. The number of different gatherings possible at the party is:

- (A)  $3^n$
- (B)  $n \times 2^n$
- (C)  $(n+1)^n$
- (D)  $2^n \times n!$

*Counting & Basic Probability · [MSQ · 2 marks]*

Q12. Which of the following are TRUE about basic counting principles?

- (A) The multiplication principle applies when a task consists of a sequence of independent choices/stages
- (B) The addition principle applies when counting the total ways to do one of several mutually exclusive alternatives
- (C) Permutations count arrangements where order matters; combinations count selections where order does not matter
- (D)  ${}^n P_r$  is always  $\geq {}^n C_r$ , for the same  $n$  and  $r \geq 1$

*Counting & Basic Probability · [NAT · 1 mark]*

Q13. Two fair dice are rolled. Find the probability that the sum of the two numbers shown is 7 (correct to 4 decimals).

*Conditional Probability & Bayes' Theorem · [MCQ · 1 mark]*

Q14. Bayes' theorem states that  $P(A|B)$  equals:

- (A)  $P(B|A)P(A) / P(B)$
- (B)  $P(A)P(B)$
- (C)  $P(A|B)P(B)$
- (D)  $P(B) / P(A)$

*Conditional Probability & Bayes' Theorem · [NAT · 2 marks]*

Q15. A disease affects 1% of a population. A diagnostic test has a 95% true positive rate (sensitivity) and a 90% true negative rate (specificity, i.e. a 10% false positive rate). If a random person tests positive, find the probability they actually have the disease (correct to 4 decimals).

*Conditional Probability & Bayes' Theorem · [NAT · 2 marks]*

Q16. A box has 2 defective and 8 good items (10 total). Two items are drawn without replacement. Given that the first item drawn is defective, find  $P(\text{second item is also defective})$ .

*Conditional Probability & Bayes' Theorem · [MSQ · 2 marks]*

Q17. Which of the following are TRUE about independent events A and B?

- (A)  $P(A \cap B) = P(A)P(B)$
- (B) Independent events are always mutually exclusive
- (C) If A and B are independent, then A and  $B^c$  (complement of B) are also independent
- (D) Independence is a property of the events' probabilities, not of the specific outcomes in the sample space

*Discrete Random Variables & PMF · [MCQ · 2 marks]*

Q18. Six unbiased dice are rolled simultaneously. Let  $X$  denote the number of distinct face values that appear among the six outcomes. What is  $P(X = 6)$ , i.e. the probability that all six numbers 1 through 6 appear exactly once?

- (A)  $5/324$
- (B)  $1/324$
- (C)  $5/1296$
- (D)  $1/6$

*Discrete Random Variables & PMF · [MSQ · 2 marks]*

Q19. Which of the following properties must every valid PMF  $p(x)$  of a discrete random variable satisfy?

- (A)  $p(x) \geq 0$  for every  $x$
- (B)  $\sum p(x) = 1$ , summed over the support
- (C)  $p(x)$  is non-decreasing as  $x$  increases
- (D)  $p(x) \leq 1$  for every individual  $x$

*Discrete Random Variables & PMF · [NAT · 1 mark]*

Q20. A discrete random variable  $X$  takes values  $\{0,1,2,3\}$  with  $P(X=0)=0.2$ ,  $P(X=1)=0.3$ ,  $P(X=2)=k$ ,  $P(X=3)=0.2$ . Find  $k$ .

*Continuous R.V. & PDF/CDF · [NAT · 2 marks]*

Q21. Let  $X$  be a continuous random variable whose CDF  $F_X(x)$ , for some  $t$ , is given by  $F_X(x) = 0$  for  $x \leq t$ ;  $(x-t)/(4-t)$  for  $t \leq x \leq 4$ ; and  $1$  for  $x \geq 4$ . If the median of  $X$  is  $3$ , find the value of  $t$ .

*Continuous R.V. & PDF/CDF · [MCQ · 1 mark]*

Q22. Which of the following is TRUE for the PDF  $f(x)$  of any continuous random variable  $X$ ?

- (A)  $f(x)$  is always  $\leq 1$  for all  $x$
- (B)  $f(x)$  directly represents  $P(X=x)$
- (C)  $\int f(x)dx$  over the entire real line equals  $1$ , and  $f(x) \geq 0$  everywhere
- (D)  $f(x)$  must be a strictly decreasing function

*Continuous R.V. & PDF/CDF · [MCQ · 1 mark]*

Q23. For  $X \sim \text{Uniform}(a,b)$ , the variance is:

- (A)  $(b-a)/2$
- (B)  $(b-a)^2/12$
- (C)  $(b-a)^2/4$
- (D)  $(b+a)/2$

*Expectation, Variance, Moments · [MCQ · 1 mark]*

Q24. Suppose  $X$  and  $Y$  are random variables. The conditional expectation of  $X$  given  $Y$  is denoted  $E[X|Y]$ . Then  $E[E[X|Y]]$  equals:

- (A)  $E[XY]$
- (B)  $E[Y]/E[X]$
- (C)  $E[X]$
- (D)  $E[Y]$

*Expectation, Variance, Moments · [NAT · 2 marks]*

Q25.  $X$  is an exponentially distributed random variable with rate  $\lambda$ . It is given that  $5 \cdot E(X) = \text{Var}(X)$ . Find the value of  $\lambda$  (correct to 1 decimal).

*Expectation, Variance, Moments · [MSQ · 2 marks]*

Q26. Which of the following are TRUE regarding the covariance of two random variables  $X$  and  $Y$ ?

- (A)  $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$
- (B) If  $X$  and  $Y$  are independent,  $\text{Cov}(X, Y) = 0$
- (C)  $\text{Cov}(X, Y) = 0$  always implies  $X$  and  $Y$  are independent
- (D)  $\text{Cov}(X, X) = \text{Var}(X)$

*Standard Distributions I · [NAT · 1 mark]*

Q27. A family has three children, where each child independently has probability  $1/2$  of being a boy and  $1/2$  of being a girl. Using the binomial PMF  $P(X=k) = C(n, k)p^k(1-p)^{n-k}$ , find the probability that the family has exactly 2 girls.

*Standard Distributions I · [MCQ · 1 mark]*

Q28. Consider the following statements: (i) The mean and variance of a Poisson random variable are equal. (ii) For a standard normal random variable, the mean is zero and the variance is one. Which ONE of the following options is correct?

- (A) Both (i) and (ii) are true
- (B) (i) is true and (ii) is false
- (C) (ii) is true and (i) is false
- (D) Both (i) and (ii) are false

*Standard Distributions I · [NAT · 1 mark]*

Q29.  $X \sim \text{Binomial}(n=8, p=0.5)$ . Find  $E[X]$ .

## Track B — Machine Learning (Q30-Q47)

*Intro to ML, Bias-Variance · [MCQ · 2 marks]*

Q30. Increasing the regularization coefficient value for a Ridge regressor will: (i) Increase or maintain model bias. (ii) Decrease model bias. (iii) Increase or maintain model variance. (iv) Decrease model variance. Which combination is correct?

- (A) i and iii
- (B) i and iv
- (C) ii and iii
- (D) ii and iv

*Intro to ML, Bias-Variance · [MCQ · 1 mark]*

Q31. The bias-variance tradeoff describes the relationship between:

- (A) Model training time and testing time
- (B) A model's error from overly simplistic assumptions (bias) and its error from sensitivity to the specific training data (variance)
- (C) The number of features and the number of training examples
- (D) The learning rate and the number of iterations

*Intro to ML, Bias-Variance · [MSQ · 2 marks]*

Q32. Which of the following statements about overfitting are TRUE?

- (A) An overfit model performs very well on training data but poorly on unseen test data

- (B) Overfitting is associated with high variance
- (C) Overfitting is associated with low variance
- (D) Regularization is a common technique used to reduce overfitting

*Linear Regression (simple) · [MCQ · 1 mark]*

Q33. In simple linear regression **without an intercept**, i.e.  $\hat{y} = w \cdot x$ , fitted to  $m$  data points  $(x_i, y_i)$  by minimizing the sum of squared errors, the optimal value of  $w$  is:

- (A)  $w = \sum x_i y_i / \sum x_i^2$
- (B)  $w = \sum x_i y_i / \sum y_i^2$
- (C)  $w = (\sum x_i)(\sum y_i) / m$
- (D)  $w = \sum x_i^2 / \sum x_i y_i$

*Linear Regression (simple) · [NAT · 2 marks]*

Q34. Given data points (1,2), (2,3), (3,5), (4,4), fit  $y = w_0 + w_1 x$  by least squares. Find the slope  $w_1$  (correct to 2 decimals).

*Linear Regression (simple) · [MSQ · 2 marks]*

Q35. Which of the following statements about the closed-form (normal-equation-style) solution for a **simple** (single-feature) linear regression are TRUE?

- (A) It always requires inverting a large matrix, just as in multi-feature regression
- (B) It exists and is unique whenever the  $x_i$  are not all identical
- (C) It can be derived by setting the partial derivatives of the cost function to zero
- (D) It requires iterative parameter updates like gradient descent

*Linear Regression (multiple) · [MCQ · 1 mark]*

Q36. For multiple linear regression  $y = Xw$  ( $X$  is the  $m \times n$  design matrix, including an intercept column;  $w$  is the  $n \times 1$  parameter vector), the least-squares normal equation solution is:

- (A)  $w = (X^T X)^{-1} X^T y$
- (B)  $w = (X X^T)^{-1} X^T y$
- (C)  $w = X^T (X X^T)^{-1} y$
- (D)  $w = X^{-1} y$

*Linear Regression (multiple) · [MCQ · 2 marks]*

Q37. For the cost function  $J(w) = (1/2m) \|Xw - y\|^2$ , the gradient descent update rule for multiple linear regression is:

- (A)  $w := w - \alpha(1/m) X^T (Xw - y)$
- (B)  $w := w - \alpha(1/m) X (Xw - y)$
- (C)  $w := w + \alpha(1/m) X^T (Xw - y)$
- (D)  $w := w - \alpha X^T X w$

*Linear Regression (multiple) · [MSQ · 2 marks]*

Q38. Which of the following are TRUE when comparing gradient descent and the normal equation for multiple linear regression?

- (A) The normal equation gives an exact closed-form solution in a single computation (no iterations)
- (B) Gradient descent requires choosing a learning rate  $\alpha$
- (C) The normal equation remains computationally cheap even when the number of features  $n$  is very large
- (D) Gradient descent can still be applied even when  $X^T X$  is singular or ill-conditioned

*Regularization: Ridge & Lasso · [MCQ · 1 mark]*

Q39. The Ridge regression cost function is:

- (A)  $\sum (y_{xii} - \hat{y}_{xii})^2 + \lambda \sum |w_j|$
- (B)  $\sum (y_{xii} - \hat{y}_{xii})^2 + \lambda \sum w_j^2$
- (C)  $\sum |y_{xii} - \hat{y}_{xii}| + \lambda \sum w_j^2$
- (D)  $\sum (y_{xii} - \hat{y}_{xii})^2$  only, with no penalty term

*Regularization: Ridge & Lasso · [MCQ · 1 mark]*

Q40. Which regularization technique is more likely to produce a sparse model (some weights exactly zero)?

- (A) Ridge (L2)
- (B) Lasso (L1)
- (C) Both equally
- (D) Neither

*Regularization: Ridge & Lasso · [MSQ · 2 marks]*

Q41. Which of the following are TRUE about Lasso regression?

- (A) It can perform feature selection by setting some coefficients exactly to zero
- (B) Its L1 penalty term is differentiable everywhere, including at  $w=0$
- (C) It uses a penalty of the form  $\lambda \sum |w_j|$
- (D) As  $\lambda \rightarrow 0$ , Lasso reduces to ordinary least squares

*Logistic Regression · [MCQ · 1 mark]*

Q42. The sigmoid (logistic) function  $\sigma(z)$  is defined as:

- (A)  $\sigma(z) = 1/(1+e^{-z})$
- (B)  $\sigma(z) = e^z/(1+z)$
- (C)  $\sigma(z) = z/(1+z^2)$
- (D)  $\sigma(z) = 1/(1-e^{-z})$

*Logistic Regression · [MCQ · 1 mark]*

Q43. In logistic regression, the decision boundary (for a classification threshold of 0.5) is defined by:

- (A)  $w^T x + b = 0.5$
- (B)  $w^T x + b = 0$
- (C)  $w^T x + b = 1$
- (D)  $\sigma(w^T x + b) = 1$

*Logistic Regression · [MSQ · 2 marks]*

Q44. Which of the following are TRUE about the sigmoid function used in logistic regression?

- (A) Its output always lies strictly between 0 and 1
- (B) It is monotonically increasing
- (C) Its derivative can be written as  $\sigma(z)(1-\sigma(z))$
- (D) It maps large negative inputs to values close to 1

*Classification Metrics · [NAT · 1 mark]*

Q45. A binary classifier produces the confusion matrix: TP=40, FP=10, FN=20, TN=30. Find the Precision.

*Classification Metrics · [NAT · 2 marks]*

Q46. Using the same confusion matrix, find the F1 score (correct to 4 decimals).

*Classification Metrics · [MSQ · 2 marks]*

Q47. Which of the following are TRUE about classification metrics?

- (A) High precision means few false positives, relative to predicted positives
- (B) High recall means few false negatives, relative to actual positives
- (C) F1 score is the harmonic mean of precision and recall
- (D) Accuracy is always the best metric to use, regardless of class balance

## Track C — Database Management Systems (Q48-Q65)

*ER Model · [MCQ · 1 mark]*

Q48. Which of the following is INCORRECT about an ER Model?

- (A) An attribute of an entity can have more than one value
- (B) An attribute of an entity can be composite
- (C) In a row of a relational table, an attribute can have more than one value
- (D) In a row of a relational table, an attribute can have exactly one value, or NULL

*ER Model · [NAT · 1 mark]*

Q49. An ER diagram has 3 strong entity sets and 2 weak entity sets. If each entity set (strong or weak) maps to exactly one table in the relational schema (ignoring relationship tables for this count), how many entity tables are needed in total?

*ER Model · [MSQ · 2 marks]*

Q50. Which of the following are valid types of cardinality ratios (mapping cardinalities) for a binary relationship?

- (A) One-to-one (1:1)
- (B) One-to-many (1:m)
- (C) Many-to-many (m:n)
- (D) Zero-to-zero (0:0)

*EER + ER-to-Relational Mapping · [MCQ · 1 mark]*

Q51. In Enhanced ER (EER) modelling, "specialization" refers to the process of:

- (A) Merging two entity sets into one
- (B) A top-down process: identifying subgroups (subclasses) within an entity set (superclass) that have distinguishing characteristics
- (C) Converting a weak entity into a strong entity
- (D) Removing redundant attributes

*EER + ER-to-Relational Mapping · [MCQ · 2 marks]*

Q52. In EER specialization/generalization, the "disjoint" vs "overlapping" constraint specifies:

- (A) Whether the subclasses are stored in separate physical files
- (B) Whether an entity can simultaneously belong to more than one subclass (overlapping allows this; disjoint does not)
- (C) Whether the relationship is 1:1 or 1:m
- (D) Whether attributes are composite or simple

*EER + ER-to-Relational Mapping · [MSQ · 2 marks]*

Q53. Which of the following are TRUE about mapping EER specialization/generalization hierarchies to relational schemas?

- (A) One common strategy creates a separate table for the superclass, plus one table per subclass (containing only that subclass's specific attributes and the inherited primary key)

- (B) Another common strategy creates one table per subclass, each including ALL inherited superclass attributes (duplicating them), with no separate superclass table
- (C) The choice of mapping strategy can depend on whether the specialization is total or partial, and disjoint or overlapping
- (D) EER hierarchies can never be mapped to a relational schema at all

*Relational Model - Keys & Constraints · [MCQ · 2 marks]*

Q54. Let  $R(a,b,c)$  and  $S(d,e,f)$  be two relations in which  $d$  is a foreign key of  $S$  that refers to the primary key of  $R$ . Consider the following operations: I. Insert into  $R$ , II. Insert into  $S$ , III. Delete from  $R$ , IV. Delete from  $S$ . Which of these operations can potentially cause a violation of the referential integrity constraint above?

- (A) I and II
- (B) II and III
- (C) I and III
- (D) II and IV

*Relational Model - Keys & Constraints · [MCQ · 1 mark]*

Q55. A candidate key of a relation is defined as:

- (A) Any attribute whose values happen to all be distinct
- (B) A minimal superkey (a superkey with no proper subset that is itself a superkey)
- (C) The primary key selected by the database administrator
- (D) Any set of attributes referencing another table

*Relational Model - Keys & Constraints · [MCQ · 1 mark]*

Q56. The "ON DELETE CASCADE" referential action means that when a referenced (parent) row is deleted:

- (A) The delete operation is rejected
- (B) All matching child rows are automatically deleted as well
- (C) The foreign key value in matching child rows is set to NULL
- (D) Nothing happens to the child rows

*Relational Algebra - Operations · [MCQ · 2 marks]*

Q57. The relation schema  $Person(pid, city)$  describes the city of residence for every person, uniquely identified by  $pid$ . Using only selection, projection, cross product, and rename, what is the minimum number of cross-product operations needed to find the list of cities where at least 3 persons reside?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

*Relational Algebra - Operations · [NAT · 2 marks]*

Q58. Relations  $r_1(P,Q,R)$  and  $r_2(R,S,T)$  have primary keys  $P$  and  $R$  respectively.  $r_1$  contains 2000 tuples and  $r_2$  contains 2500 tuples. What is the maximum possible number of tuples in the natural join  $r_1 \bowtie r_2$  (joined on the common attribute  $R$ )?

*Relational Algebra - Operations · [MSQ · 2 marks]*

Q59. Which of the following are considered fundamental (primitive) operations of relational algebra, from which other operations (like natural join) can be derived?

- (A) Selection ( $\sigma$ )
- (B) Projection ( $\pi$ )

- (C) Union ( $\cup$ )
- (D) Natural Join ( $\bowtie$ )

*Relational Calculus (Tuple & Domain) · [MCQ · 2 marks]*

Q60. Which of the following relational query languages have the same expressive power? I) Relational algebra. II) Tuple relational calculus restricted to safe expressions. III) Domain relational calculus restricted to safe expressions.

- (A) II and III only
- (B) I and II only
- (C) I and III only
- (D) I, II and III

*Relational Calculus (Tuple & Domain) · [MCQ · 1 mark]*

Q61. Which of the following tuple relational calculus expressions is equivalent to  $\forall t \in r (P(t))$ ?

- (A)  $\neg \exists t \in r (\neg P(t))$
- (B)  $\exists t \in r (\neg P(t))$
- (C)  $\neg \exists t \in r (P(t))$
- (D)  $\exists t \notin r (P(t))$

*Relational Calculus (Tuple & Domain) · [NAT · 1 mark]*

Q62. Relation R(A,B,C) has 100 tuples. A TRC query  $\{t.A \mid R(t) \wedge t.B = 5\}$  finds 15 matching tuples. How many tuples does the result relation contain (assuming all resulting A-values happen to be distinct)?

*SQL Basics · [MCQ · 1 mark]*

Q63. The SELECT operation in SQL is equivalent to which relational algebra operation?

- (A) The selection operation in relational algebra
- (B) The selection operation in relational algebra, except that SQL's SELECT retains duplicate rows
- (C) The projection operation in relational algebra
- (D) The projection operation in relational algebra, except that SQL's SELECT retains duplicate rows

*SQL Basics · [MCQ · 2 marks]*

Q64. Given schemas employees(emp-id, first-name, last-name, hire-date, dept-id, salary) and departments(dept-id, dept-name, manager-id, location-id), to display the last names and hire dates of the MOST RECENTLY HIRED employee in each department at location-id 1700, which SQL technique is essential?

- (A) A simple WHERE clause only, with no subquery
- (B) A subquery using MAX(hire-date) combined with GROUP BY dept-id, matched via IN against (dept-id, hire-date) pairs
- (C) ORDER BY hire-date DESC alone, with no grouping
- (D) DISTINCT alone, with no subquery

*SQL Basics · [MSQ · 2 marks]*

Q65. Which of the following are valid uses of the WHERE clause?

- (A) Filtering rows based on a numeric comparison, e.g. WHERE age > 18
- (B) Filtering rows based on a string pattern using LIKE
- (C) Combining multiple conditions using AND / OR
- (D) Filtering aggregated group results, e.g. WHERE COUNT(\*) > 5

# Answer Key

| Q# | Type | Marks | Answer             | Q# | Type | Marks | Answer        | Q# | Type | Marks | Answer        |
|----|------|-------|--------------------|----|------|-------|---------------|----|------|-------|---------------|
| 1  | NAT  | 1     | 4800               | 23 | MCQ  | 1     | (B)           | 45 | NAT  | 1     | 0.8           |
| 2  | MCQ  | 2     | (B)                | 24 | MCQ  | 1     | (C)           | 46 | NAT  | 2     | 0.7273        |
| 3  | NAT  | 1     | 40                 | 25 | NAT  | 2     | 0.2           | 47 | MSQ  | 2     | (A), (B), (C) |
| 4  | MCQ  | 2     | (A)                | 26 | MSQ  | 2     | (A), (B), (D) | 48 | MCQ  | 1     | (C)           |
| 5  | NAT  | 1     | 5.5                | 27 | NAT  | 1     | 0.375         | 49 | NAT  | 1     | 5             |
| 6  | MCQ  | 2     | (A)                | 28 | MCQ  | 1     | (A)           | 50 | MSQ  | 2     | (A), (B), (C) |
| 7  | NAT  | 1     | 25                 | 29 | NAT  | 1     | 4             | 51 | MCQ  | 1     | (B)           |
| 8  | MCQ  | 2     | (A)                | 30 | MCQ  | 2     | (B)           | 52 | MCQ  | 2     | (B)           |
| 9  | NAT  | 1     | 6                  | 31 | MCQ  | 1     | (B)           | 53 | MSQ  | 2     | (A), (B), (C) |
| 10 | MCQ  | 2     | (A)                | 32 | MSQ  | 2     | (A), (B), (D) | 54 | MCQ  | 2     | (B)           |
| 11 | MCQ  | 2     | (A)                | 33 | MCQ  | 1     | (A)           | 55 | MCQ  | 1     | (B)           |
| 12 | MSQ  | 2     | (A), (B), (C), (D) | 34 | NAT  | 2     | 0.80          | 56 | MCQ  | 1     | (B)           |
| 13 | NAT  | 1     | 0.1667             | 35 | MSQ  | 2     | (B), (C)      | 57 | MCQ  | 2     | (B)           |
| 14 | MCQ  | 1     | (A)                | 36 | MCQ  | 1     | (A)           | 58 | NAT  | 2     | 2000          |
| 15 | NAT  | 2     | 0.0876             | 37 | MCQ  | 2     | (A)           | 59 | MSQ  | 2     | (A), (B), (C) |
| 16 | NAT  | 2     | 0.1111             | 38 | MSQ  | 2     | (A), (B), (D) | 60 | MCQ  | 2     | (D)           |
| 17 | MSQ  | 2     | (A), (C), (D)      | 39 | MCQ  | 1     | (B)           | 61 | MCQ  | 1     | (A)           |
| 18 | MCQ  | 2     | (A)                | 40 | MCQ  | 1     | (B)           | 62 | NAT  | 1     | 15            |
| 19 | MSQ  | 2     | (A), (B), (D)      | 41 | MSQ  | 2     | (A), (C), (D) | 63 | MCQ  | 1     | (D)           |
| 20 | NAT  | 1     | 0.30               | 42 | MCQ  | 1     | (A)           | 64 | MCQ  | 2     | (B)           |
| 21 | NAT  | 2     | 2                  | 43 | MCQ  | 1     | (B)           | 65 | MSQ  | 2     | (A), (B), (C) |
| 22 | MCQ  | 1     | (C)                | 44 | MSQ  | 2     | (A), (B), (C) |    |      |       |               |

# Full Solutions

## Section 1: General Aptitude

[GATE-style]

### Q1. Percentages [NAT · 1 mark]

In an election between two candidates, one candidate got 60% of the total valid votes. If 8000 valid votes were cast in total, find the number of votes the winning candidate received.

**Answer: 4800**

**Solution:**  $0.6 \times 8000 = 4800$ .

[GATE-style]

### Q2. Percentages [MCQ · 2 marks]

A number is increased by 20% and then decreased by 20%. Find the net percentage change.

- (A) 0%
- (B) -4%
- (C) +4%
- (D) -20%

**Answer: (B)**

**Solution:** Let  $x=100$ . After +20%: 120. After -20% of 120 (=24):  $120-24=96$ . Net change =  $96-100 = -4\%$ .

[GATE-style]

### Q3. Ratio & Proportion [NAT · 1 mark]

In a class, the ratio of girls to boys is 3:5. If there are 24 girls, how many boys are there?

**Answer: 40**

**Solution:** girls/boys =  $3/5 \Rightarrow 24/\text{boys} = 3/5 \Rightarrow \text{boys} = 24 \times 5/3 = 40$ .

[GATE-style]

### Q4. Ratio & Proportion [MCQ · 2 marks]

$p:q = 1:2$ ,  $q:r = 4:3$ ,  $r:s = 4:5$ . If  $u$  is 50% more than  $s$ , find  $p:u$ .

- (A) 16:45
- (B) 8:15
- (C) 45:16
- (D) 15:8

**Answer: (A)**

**Solution:** Let  $s=15 \Rightarrow r = (4/5) \times 15 = 12 \Rightarrow q = (4/3) \times 12 = 16 \Rightarrow p = q/2 = 8$ .  $u = 1.5 \times 15 = 22.5$ .  
 $p:u = 8:22.5 = 16:45$ .

[GATE-style]

### Q5. Averages [NAT · 1 mark]

Find the average of the first 10 natural numbers.

**Answer: 5.5**

**Solution:** Sum = 55, count = 10, average =  $55/10 = 5.5$ .

[GATE-style]

**Q6. Averages [MCQ · 2 marks]**

The average weight of 8 people increases by 2.5 kg when a new person replaces one of them who weighed 60 kg. Find the weight of the new person.

- (A) 80 kg
- (B) 70 kg
- (C) 65 kg
- (D) 75 kg

**Answer: (A)**

**Solution:** Total weight increases by  $8 \times 2.5 = 20$  kg. So new person's weight =  $60 + 20 = 80$  kg.

---

[GATE-style]

**Q7. Profit, Loss & Discount [NAT · 1 mark]**

A shopkeeper buys an item for Rs. 400 and sells it for Rs. 500. Find the profit percentage.

**Answer: 25**

**Solution:** Profit = 100. Profit% =  $(100/400) \times 100 = 25\%$ .

---

[GATE-style]

**Q8. Profit, Loss & Discount [MCQ · 2 marks]**

A shopkeeper marks an item 40% above cost price and then gives a discount of 10% on the marked price. Find the overall profit percentage.

- (A) 26%
- (B) 30%
- (C) 25%
- (D) 24%

**Answer: (A)**

**Solution:** Let CP=100. MP = 140. SP =  $140 \times 0.9 = 126$ . Profit% =  $126 - 100 = 26\%$ .

---

[GATE-style]

**Q9. Time & Work [NAT · 1 mark]**

A can complete a work in 10 days, and B can complete the same work in 15 days. If they work together, how many days will they take to complete the work?

**Answer: 6**

**Solution:** Rate of A =  $1/10$ , rate of B =  $1/15$ . Combined rate =  $1/10 + 1/15 = 3/30 + 2/30 = 5/30 = 1/6$ .  
Time = 6 days.

---

[GATE-style]

**Q10. Time, Speed & Distance [MCQ · 2 marks]**

Two trains start from the same point and travel in OPPOSITE directions at 60 km/h and 40 km/h respectively. After how many hours will they be 300 km apart?

- (A) 3
- (B) 2
- (C) 4
- (D) 5

**Answer: (A)**

**Solution:** Relative speed (opposite directions) =  $60 + 40 = 100$  km/h. Time =  $300/100 = 3$  hours.

---

## Section 2, Track A: Probability & Statistics

[VERIFIED PYQ — GATE CSE 2003, classic counting question]

### Q11. Counting & Basic Probability [MCQ · 2 marks]

$n$  couples are invited to a party with the condition that every husband should be accompanied by his wife; however, a wife need not be accompanied by her husband. The number of different gatherings possible at the party is:

- (A)  $3^n$
- (B)  $n \times 2^n$
- (C)  $(n+1)^n$
- (D)  $2^n \times n!$

**Answer: (A)**

**Solution:** For each of the  $n$  couples, there are exactly 3 independent possibilities: neither attends, only the wife attends, or both attend (the husband can never attend alone). By the multiplication principle across  $n$  independent couples, the total number of gatherings =  $3^n$ .

[GATE-style]

### Q12. Counting & Basic Probability [MSQ · 2 marks]

Which of the following are TRUE about basic counting principles?

- (A) The multiplication principle applies when a task consists of a sequence of independent choices/stages
- (B) The addition principle applies when counting the total ways to do one of several mutually exclusive alternatives
- (C) Permutations count arrangements where order matters; combinations count selections where order does not matter
- (D)  ${}^n P_r$  is always  $\geq {}^n C_r$ , for the same  $n$  and  $r \geq 1$

**Answer: (A), (B), (C), (D)**

**Solution:** All four are standard, correct facts. (D) holds since  ${}^n P_r = {}^n C_r \times r!$ , and  $r! \geq 1$  for  $r \geq 1$ .

[GATE-style]

### Q13. Counting & Basic Probability [NAT · 1 mark]

Two fair dice are rolled. Find the probability that the sum of the two numbers shown is 7 (correct to 4 decimals).

**Answer: 0.1667**

**Solution:** Favourable outcomes: (1,6),(2,5),(3,4),(4,3),(5,2),(6,1) = 6. Total outcomes = 36.  $P = 6/36 \approx 0.1667$ .

[GATE-style]

### Q14. Conditional Probability & Bayes' Theorem [MCQ · 1 mark]

Bayes' theorem states that  $P(A|B)$  equals:

- (A)  $P(B|A)P(A) / P(B)$
- (B)  $P(A)P(B)$
- (C)  $P(A|B)P(B)$
- (D)  $P(B) / P(A)$

**Answer: (A)**

[GATE-style, classic Bayes disease-test problem]

### Q15. Conditional Probability & Bayes' Theorem [NAT · 2 marks]

A disease affects 1% of a population. A diagnostic test has a 95% true positive rate (sensitivity) and a 90% true negative rate (specificity, i.e. a 10% false positive rate). If a random person tests positive, find the probability they actually have the disease (correct to 4 decimals).

**Answer: 0.0876**

**Solution:**  $P(D)=0.01$ ,  $P(+|D)=0.95$ ,  $P(+|D')=0.10$ ,  $P(D')=0.99$ .

$P(+) = 0.95 \times 0.01 + 0.10 \times 0.99 = 0.0095 + 0.099 = 0.1085$ .

By Bayes' theorem:  $P(D|+) = P(+|D)P(D)/P(+) = 0.0095/0.1085 \approx 0.0876$ . (This illustrates why even a fairly accurate test yields a low posterior probability when the base rate of the disease is very low.)

---

[GATE-style]

**Q16. Conditional Probability & Bayes' Theorem [NAT · 2 marks]**

A box has 2 defective and 8 good items (10 total). Two items are drawn without replacement. Given that the first item drawn is defective, find  $P(\text{second item is also defective})$ .

**Answer: 0.1111**

**Solution:** After removing one defective item, 1 defective and 8 good items remain (9 total).  $P(\text{2nd defective} | \text{1st defective}) = 1/9 \approx 0.1111$ .

---

[GATE-style]

**Q17. Conditional Probability & Bayes' Theorem [MSQ · 2 marks]**

Which of the following are TRUE about independent events A and B?

- (A)  $P(A \cap B) = P(A)P(B)$
- (B) Independent events are always mutually exclusive
- (C) If A and B are independent, then A and  $B^c$  (complement of B) are also independent
- (D) Independence is a property of the events' probabilities, not of the specific outcomes in the sample space

**Answer: (A), (C), (D)**

**Solution:** (B) is false -- independence and mutual exclusivity are different (and generally incompatible, unless one event has probability 0): if A and B are mutually exclusive with positive probabilities,  $P(A \cap B) = 0 \neq P(A)P(B)$  in general, so they cannot be independent.

---

[VERIFIED PYQ — GATE CSE 2024, Set 2 (same computation, reframed with an explicit random variable)]

**Q18. Discrete Random Variables & PMF [MCQ · 2 marks]**

Six unbiased dice are rolled simultaneously. Let X denote the number of distinct face values that appear among the six outcomes. What is  $P(X = 6)$ , i.e. the probability that all six numbers 1 through 6 appear exactly once?

- (A)  $5/324$
- (B)  $1/324$
- (C)  $5/1296$
- (D)  $1/6$

**Answer: (A)**

**Solution:** Total equally-likely outcomes =  $6^6 = 46656$ . Favourable outcomes (all six faces distinct) =  $6! = 720$  (each die gets a different face -- a permutation of  $\{1, \dots, 6\}$ ).  $P(X=6) = 720/46656 = 5/324$ .

---

[GATE-style]

**Q19. Discrete Random Variables & PMF [MSQ · 2 marks]**

Which of the following properties must every valid PMF  $p(x)$  of a discrete random variable satisfy?

- (A)  $p(x) \geq 0$  for every x

- (B)  $\sum p(x) = 1$ , summed over the support
- (C)  $p(x)$  is non-decreasing as  $x$  increases
- (D)  $p(x) \leq 1$  for every individual  $x$

**Answer: (A), (B), (D)**

**Solution:** (C) is false -- there is no requirement that the PMF be monotonic in  $x$ ; it can rise and fall arbitrarily.

[GATE-style]

**Q20. Discrete Random Variables & PMF [NAT · 1 mark]**

A discrete random variable  $X$  takes values  $\{0,1,2,3\}$  with  $P(X=0)=0.2$ ,  $P(X=1)=0.3$ ,  $P(X=2)=k$ ,  $P(X=3)=0.2$ . Find  $k$ .

**Answer: 0.30**

**Solution:** Since probabilities must sum to 1:  $0.2+0.3+k+0.2 = 1 \Rightarrow k = 0.30$ .

[VERIFIED PYQ — GATE DA 2025]

**Q21. Continuous R.V. & PDF/CDF [NAT · 2 marks]**

Let  $X$  be a continuous random variable whose CDF  $F_X(x)$ , for some  $t$ , is given by  $F_X(x) = 0$  for  $x \leq t$ ;  $(x-t)/(4-t)$  for  $t \leq x \leq 4$ ; and 1 for  $x \geq 4$ . If the median of  $X$  is 3, find the value of  $t$ .

**Answer: 2**

**Solution:** At the median,  $F_X(3) = 0.5$ . So  $(3-t)/(4-t) = 0.5 \Rightarrow 3-t = 0.5(4-t) = 2-0.5t \Rightarrow 3-t-2+0.5t = 0 \Rightarrow 1-0.5t = 0 \Rightarrow t = 2$ . (Check:  $(3-2)/(4-2) = 1/2$  ✓)

[GATE-style]

**Q22. Continuous R.V. & PDF/CDF [MCQ · 1 mark]**

Which of the following is TRUE for the PDF  $f(x)$  of any continuous random variable  $X$ ?

- (A)  $f(x)$  is always  $\leq 1$  for all  $x$
- (B)  $f(x)$  directly represents  $P(X=x)$
- (C)  $\int f(x)dx$  over the entire real line equals 1, and  $f(x) \geq 0$  everywhere
- (D)  $f(x)$  must be a strictly decreasing function

**Answer: (C)**

[New Practice]

**Q23. Continuous R.V. & PDF/CDF [MCQ · 1 mark]**

For  $X \sim \text{Uniform}(a,b)$ , the variance is:

- (A)  $(b-a)/2$
- (B)  $(b-a)^2/12$
- (C)  $(b-a)^2/4$
- (D)  $(b+a)/2$

**Answer: (B)**

[VERIFIED PYQ — GATE DA 2025, Question 11]

**Q24. Expectation, Variance, Moments [MCQ · 1 mark]**

Suppose  $X$  and  $Y$  are random variables. The conditional expectation of  $X$  given  $Y$  is denoted  $E[X|Y]$ . Then  $E[E[X|Y]]$  equals:

- (A)  $E[XY]$
- (B)  $E[Y]/E[X]$
- (C)  $E[X]$

(D)  $E[Y]$

**Answer: (C)**

**Solution:** This is the Law of Total Expectation (law of iterated expectations):  $E[E[X|Y]] = E[X]$ , for any random variables  $X$  and  $Y$ .

---

[VERIFIED PYQ — GATE DA 2024]

**Q25. Expectation, Variance, Moments [NAT · 2 marks]**

$X$  is an exponentially distributed random variable with rate  $\lambda$ . It is given that  $5 \cdot E(X) = \text{Var}(X)$ . Find the value of  $\lambda$  (correct to 1 decimal).

**Answer: 0.2**

**Solution:** For  $\text{Exponential}(\lambda)$ :  $E(X) = 1/\lambda$ ,  $\text{Var}(X) = 1/\lambda^2$ . So  $5(1/\lambda) = 1/\lambda^2 \Rightarrow 5\lambda = 1 \Rightarrow \lambda = 0.2$ .

---

[New Practice]

**Q26. Expectation, Variance, Moments [MSQ · 2 marks]**

Which of the following are TRUE regarding the covariance of two random variables  $X$  and  $Y$ ?

- (A)  $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$
- (B) If  $X$  and  $Y$  are independent,  $\text{Cov}(X, Y) = 0$
- (C)  $\text{Cov}(X, Y) = 0$  always implies  $X$  and  $Y$  are independent
- (D)  $\text{Cov}(X, X) = \text{Var}(X)$

**Answer: (A), (B), (D)**

**Solution:** (C) is false -- zero covariance means no LINEAR relationship, but  $X$  and  $Y$  can still be dependent through a non-linear relationship (e.g.  $Y=X^2$  for symmetric  $X$  can have zero covariance yet clear dependence).

---

[VERIFIED PYQ — GATE DA 2024, Question 7]

**Q27. Standard Distributions I [NAT · 1 mark]**

A family has three children, where each child independently has probability  $1/2$  of being a boy and  $1/2$  of being a girl. Using the binomial PMF  $P(X=k) = C(n, k)p^k(1-p)^{n-k}$ , find the probability that the family has exactly 2 girls.

**Answer: 0.375**

**Solution:**  $n=3$ ,  $p=1/2$ ,  $k=2$ .  $P(2 \text{ girls}) = C(3, 2)(1/2)^2(1/2)^1 = 3 \times (1/4) \times (1/2) = 3/8 = 0.375$ .

---

[VERIFIED PYQ — GATE DA 2024, Question 11]

**Q28. Standard Distributions I [MCQ · 1 mark]**

Consider the following statements: (i) The mean and variance of a Poisson random variable are equal. (ii) For a standard normal random variable, the mean is zero and the variance is one. Which ONE of the following options is correct?

- (A) Both (i) and (ii) are true
- (B) (i) is true and (ii) is false
- (C) (ii) is true and (i) is false
- (D) Both (i) and (ii) are false

**Answer: (A)**

**Solution:** For  $\text{Poisson}(\lambda)$ , both the mean and variance equal  $\lambda$  -- statement (i) is true. For the standard normal distribution  $N(0, 1)$ , mean=0 and variance=1 by definition -- statement (ii) is also true. So both statements are true.

---

[New Practice]

**Q29. Standard Distributions I [NAT · 1 mark]**

$X \sim \text{Binomial}(n=8, p=0.5)$ . Find  $E[X]$ .

**Answer: 4**

**Solution:**  $E[X] = np = 8 \times 0.5 = 4$ .

## Section 2, Track B: Machine Learning

[VERIFIED PYQ — GATE DA 2024, Official Sample Question Paper, Question 12]

**Q30. Intro to ML, Bias-Variance [MCQ · 2 marks]**

Increasing the regularization coefficient value for a Ridge regressor will: (i) Increase or maintain model bias. (ii) Decrease model bias. (iii) Increase or maintain model variance. (iv) Decrease model variance. Which combination is correct?

- (A) i and iii
- (B) i and iv
- (C) ii and iii
- (D) ii and iv

**Answer: (B)**

**Solution:** A larger regularization coefficient pushes the model toward simpler solutions (smaller weight magnitudes). This increases (or at least maintains) bias (i), while making the model less sensitive to the specific training sample, decreasing variance (iv) -- a direct illustration of the bias-variance tradeoff.

[GATE-style]

**Q31. Intro to ML, Bias-Variance [MCQ · 1 mark]**

The bias-variance tradeoff describes the relationship between:

- (A) Model training time and testing time
- (B) A model's error from overly simplistic assumptions (bias) and its error from sensitivity to the specific training data (variance)
- (C) The number of features and the number of training examples
- (D) The learning rate and the number of iterations

**Answer: (B)**

[New Practice]

**Q32. Intro to ML, Bias-Variance [MSQ · 2 marks]**

Which of the following statements about overfitting are TRUE?

- (A) An overfit model performs very well on training data but poorly on unseen test data
- (B) Overfitting is associated with high variance
- (C) Overfitting is associated with low variance
- (D) Regularization is a common technique used to reduce overfitting

**Answer: (A), (B), (D)**

[GATE-style -- concept confirmed tested in GATE DA 2025]

**Q33. Linear Regression (simple) [MCQ · 1 mark]**

In simple linear regression **without an intercept**, i.e.  $\hat{y} = w \cdot x$ , fitted to  $m$  data points  $(x_i, y_i)$  by minimizing the sum of squared errors, the optimal value of  $w$  is:

- (A)  $w = \sum x_i y_i / \sum x_i^2$
- (B)  $w = \sum x_i y_i / \sum y_i^2$
- (C)  $w = (\sum x_i)(\sum y_i) / m$

(D)  $w = \sum x_i^2 / \sum x_i y_i$

**Answer: (A)**

**Solution:**  $J(w) = \sum (y_i - wx_i)^2$ . Setting  $dJ/dw = -2\sum x_i(y_i - wx_i) = 0$  gives  $\sum x_i y_i = w \sum x_i^2$ , so  $w = \sum x_i y_i / \sum x_i^2$ . (This exact formula-recall question format matches the concept tested in GATE DA 2025.)

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[GATE-style]

**Q34. Linear Regression (simple) [NAT · 2 marks]**

Given data points (1,2), (2,3), (3,5), (4,4), fit  $y = w_0 + w_1 x$  by least squares. Find the slope  $w_1$  (correct to 2 decimals).

**Answer: 0.80**

**Solution:**  $\bar{x}=2.5$ ,  $\bar{y}=3.5$ . Deviations  $(x-\bar{x}, y-\bar{y})$ :  $(-1.5, -1.5), (-0.5, -0.5), (0.5, 1.5), (1.5, 0.5)$ .

$$\sum (x-\bar{x})(y-\bar{y}) = 2.25 + 0.25 + 0.75 + 0.75 = 4.$$

$$\sum (x-\bar{x})^2 = 2.25 + 0.25 + 0.25 + 2.25 = 5.$$

$$w_1 = 4/5 = 0.80.$$

---

[New Practice]

**Q35. Linear Regression (simple) [MSQ · 2 marks]**

Which of the following statements about the closed-form (normal-equation-style) solution for a **simple** (single-feature) linear regression are TRUE?

- (A) It always requires inverting a large matrix, just as in multi-feature regression
- (B) It exists and is unique whenever the  $x_i$  are not all identical
- (C) It can be derived by setting the partial derivatives of the cost function to zero
- (D) It requires iterative parameter updates like gradient descent

**Answer: (B), (C)**

**Solution:** For one feature,  $w_0, w_1$  have simple direct formulas (no matrix inversion needed) -- (A) is false. (D) describes gradient descent, not the closed form -- false. (B) and (C) correctly describe the normal-equation approach.

---

[GATE-style]

**Q36. Linear Regression (multiple) [MCQ · 1 mark]**

For multiple linear regression  $y = Xw$  ( $X$  is the  $m \times n$  design matrix, including an intercept column;  $w$  is the  $n \times 1$  parameter vector), the least-squares normal equation solution is:

- (A)  $w = (X^T X)^{-1} X^T y$
- (B)  $w = (X X^T)^{-1} X^T y$
- (C)  $w = X^T (X X^T)^{-1} y$
- (D)  $w = X^{-1} y$

**Answer: (A)**

**Solution:** Setting the gradient of  $J(w) = (1/2m) \|Xw - y\|^2$  to zero gives  $X^T X w = X^T y$ , so  $w = (X^T X)^{-1} X^T y$ , provided  $X^T X$  is invertible.

---

[GATE-style]

**Q37. Linear Regression (multiple) [MCQ · 2 marks]**

For the cost function  $J(w) = (1/2m) \|Xw - y\|^2$ , the gradient descent update rule for multiple linear regression is:

- (A)  $w := w - \alpha(1/m) X^T (Xw - y)$
- (B)  $w := w - \alpha(1/m) X (Xw - y)$
- (C)  $w := w + \alpha(1/m) X^T (Xw - y)$

(D)  $w := w - \alpha X^T X w$

**Answer: (A)**

**Solution:**  $\nabla J(w) = (1/m)X^T(Xw - y)$ . Gradient descent moves against the gradient:  $w := w - \alpha \nabla J(w)$ .

---

[GATE-style]

**Q38. Linear Regression (multiple) [MSQ · 2 marks]**

Which of the following are TRUE when comparing gradient descent and the normal equation for multiple linear regression?

- (A) The normal equation gives an exact closed-form solution in a single computation (no iterations)
- (B) Gradient descent requires choosing a learning rate  $\alpha$
- (C) The normal equation remains computationally cheap even when the number of features  $n$  is very large
- (D) Gradient descent can still be applied even when  $X^T X$  is singular or ill-conditioned

**Answer: (A), (B), (D)**

**Solution:** (C) is false -- computing  $(X^T X)^{-1}$  costs  $O(n^3)$ , which becomes expensive as  $n$  grows large; gradient descent scales much better in that regime.

---

[GATE-style]

**Q39. Regularization: Ridge & Lasso [MCQ · 1 mark]**

The Ridge regression cost function is:

- (A)  $\sum (y_{xii} - \hat{y}_{xii})^2 + \lambda \sum |w_j|$
- (B)  $\sum (y_{xii} - \hat{y}_{xii})^2 + \lambda \sum w_j^2$
- (C)  $\sum |y_{xii} - \hat{y}_{xii}| + \lambda \sum w_j^2$
- (D)  $\sum (y_{xii} - \hat{y}_{xii})^2$  only, with no penalty term

**Answer: (B)**

---

[GATE-style]

**Q40. Regularization: Ridge & Lasso [MCQ · 1 mark]**

Which regularization technique is more likely to produce a sparse model (some weights exactly zero)?

- (A) Ridge (L2)
- (B) Lasso (L1)
- (C) Both equally
- (D) Neither

**Answer: (B)**

---

[New Practice]

**Q41. Regularization: Ridge & Lasso [MSQ · 2 marks]**

Which of the following are TRUE about Lasso regression?

- (A) It can perform feature selection by setting some coefficients exactly to zero
- (B) Its L1 penalty term is differentiable everywhere, including at  $w=0$
- (C) It uses a penalty of the form  $\lambda \sum |w_j|$
- (D) As  $\lambda \rightarrow 0$ , Lasso reduces to ordinary least squares

**Answer: (A), (C), (D)**

**Solution:** (B) is false -- the L1 penalty  $|w|$  has a non-differentiable "kink" exactly at  $w=0$ , and this kink is precisely what allows Lasso to drive coefficients to exact zero.

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[GATE-style]

**Q42. Logistic Regression [MCQ · 1 mark]**

The sigmoid (logistic) function  $\sigma(z)$  is defined as:

- (A)  $\sigma(z) = 1/(1+e^{-z})$
- (B)  $\sigma(z) = e^z/(1+z)$
- (C)  $\sigma(z) = z/(1+z^2)$
- (D)  $\sigma(z) = 1/(1-e^{-z})$

**Answer: (A)**

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[GATE-style]

**Q43. Logistic Regression [MCQ · 1 mark]**

In logistic regression, the decision boundary (for a classification threshold of 0.5) is defined by:

- (A)  $w^T x + b = 0.5$
- (B)  $w^T x + b = 0$
- (C)  $w^T x + b = 1$
- (D)  $\sigma(w^T x + b) = 1$

**Answer: (B)**

**Solution:**  $\sigma(z)=0.5$  exactly when  $z=0$ , so the decision boundary is the set of points where  $w^T x + b = 0$ .

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[GATE-style]

**Q44. Logistic Regression [MSQ · 2 marks]**

Which of the following are TRUE about the sigmoid function used in logistic regression?

- (A) Its output always lies strictly between 0 and 1
- (B) It is monotonically increasing
- (C) Its derivative can be written as  $\sigma(z)(1-\sigma(z))$
- (D) It maps large negative inputs to values close to 1

**Answer: (A), (B), (C)**

**Solution:** (D) is false -- large negative  $z$  maps close to 0, not 1.

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[GATE-style]

**Q45. Classification Metrics [NAT · 1 mark]**

A binary classifier produces the confusion matrix: TP=40, FP=10, FN=20, TN=30. Find the Precision.

**Answer: 0.8**

**Solution:** Precision =  $40/(40+10) = 40/50 = 0.8$ .

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[GATE-style]

**Q46. Classification Metrics [NAT · 2 marks]**

Using the same confusion matrix, find the F1 score (correct to 4 decimals).

**Answer: 0.7273**

**Solution:**  $F1 = 2 \times P \times R / (P + R) = 2 \times 0.8 \times 0.6667 / (0.8 + 0.6667) = 1.0667 / 1.4667 \approx 0.7273$  (exactly 8/11).

---

[GATE-style]

**Q47. Classification Metrics [MSQ · 2 marks]**

Which of the following are TRUE about classification metrics?

- (A) High precision means few false positives, relative to predicted positives
- (B) High recall means few false negatives, relative to actual positives

- (C) F1 score is the harmonic mean of precision and recall
- (D) Accuracy is always the best metric to use, regardless of class balance

**Answer: (A), (B), (C)**

**Solution:** (D) is false -- accuracy can be misleading on imbalanced datasets (e.g. a model predicting only the majority class can have high accuracy but be useless).

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## Section 2, Track C: Database Management Systems

[VERIFIED PYQ — GATE CSE 2012]

### Q48. ER Model [MCQ · 1 mark]

Which of the following is INCORRECT about an ER Model?

- (A) An attribute of an entity can have more than one value
- (B) An attribute of an entity can be composite
- (C) In a row of a relational table, an attribute can have more than one value
- (D) In a row of a relational table, an attribute can have exactly one value, or NULL

**Answer: (C)**

**Solution:** Statement (C) is the INCORRECT one -- in the relational model, every attribute value in a row must be atomic (a single value, or NULL); this is precisely the requirement of First Normal Form. Multi-valued and composite attributes are allowed conceptually in the ER model (A, B are correct statements about the ER model), but once mapped to the relational model, values must be atomic (D is also a correct statement).

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[New Practice]

### Q49. ER Model [NAT · 1 mark]

An ER diagram has 3 strong entity sets and 2 weak entity sets. If each entity set (strong or weak) maps to exactly one table in the relational schema (ignoring relationship tables for this count), how many entity tables are needed in total?

**Answer: 5**

**Solution:** 3 (strong) + 2 (weak) = 5 entity tables.

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[New Practice]

### Q50. ER Model [MSQ · 2 marks]

Which of the following are valid types of cardinality ratios (mapping cardinalities) for a binary relationship?

- (A) One-to-one (1:1)
- (B) One-to-many (1:m)
- (C) Many-to-many (m:n)
- (D) Zero-to-zero (0:0)

**Answer: (A), (B), (C)**

**Solution:** (D) "Zero-to-zero" is not a standard cardinality ratio type.

---

[GATE-style]

### Q51. EER + ER-to-Relational Mapping [MCQ · 1 mark]

In Enhanced ER (EER) modelling, "specialization" refers to the process of:

- (A) Merging two entity sets into one
- (B) A top-down process: identifying subgroups (subclasses) within an entity set (superclass) that have distinguishing characteristics

- (C) Converting a weak entity into a strong entity
- (D) Removing redundant attributes

**Answer: (B)**

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[GATE-style]

**Q52. EER + ER-to-Relational Mapping [MCQ · 2 marks]**

In EER specialization/generalization, the "disjoint" vs "overlapping" constraint specifies:

- (A) Whether the subclasses are stored in separate physical files
- (B) Whether an entity can simultaneously belong to more than one subclass (overlapping allows this; disjoint does not)
- (C) Whether the relationship is 1:1 or 1:m
- (D) Whether attributes are composite or simple

**Answer: (B)**

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[GATE-style]

**Q53. EER + ER-to-Relational Mapping [MSQ · 2 marks]**

Which of the following are TRUE about mapping EER specialization/generalization hierarchies to relational schemas?

- (A) One common strategy creates a separate table for the superclass, plus one table per subclass (containing only that subclass's specific attributes and the inherited primary key)
- (B) Another common strategy creates one table per subclass, each including ALL inherited superclass attributes (duplicating them), with no separate superclass table
- (C) The choice of mapping strategy can depend on whether the specialization is total or partial, and disjoint or overlapping
- (D) EER hierarchies can never be mapped to a relational schema at all

**Answer: (A), (B), (C)**

**Solution:** (D) is false -- multiple well-established strategies exist precisely for mapping EER specialization/generalization to relations.

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[VERIFIED PYQ — GATE CS, classic referential-integrity question]

**Q54. Relational Model - Keys & Constraints [MCQ · 2 marks]**

Let R(a,b,c) and S(d,e,f) be two relations in which d is a foreign key of S that refers to the primary key of R. Consider the following operations: I. Insert into R, II. Insert into S, III. Delete from R, IV. Delete from S. Which of these operations can potentially cause a violation of the referential integrity constraint above?

- (A) I and II
- (B) II and III
- (C) I and III
- (D) II and IV

**Answer: (B)**

**Solution:** Inserting into R (I) only adds new primary-key values -- it cannot break any existing reference. Deleting from S (IV) only removes references -- it cannot create a dangling reference either. But inserting into S (II) with a foreign-key value not present in R violates referential integrity, and deleting a row from R (III) that is still referenced by some row in S also violates it (unless a cascade/set-null action is defined). So the answer is II and III.

---

[GATE-style]

**Q55. Relational Model - Keys & Constraints [MCQ · 1 mark]**

A candidate key of a relation is defined as:

- (A) Any attribute whose values happen to all be distinct
- (B) A minimal superkey (a superkey with no proper subset that is itself a superkey)
- (C) The primary key selected by the database administrator
- (D) Any set of attributes referencing another table

**Answer: (B)**

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[New Practice]

**Q56. Relational Model - Keys & Constraints [MCQ · 1 mark]**

The "ON DELETE CASCADE" referential action means that when a referenced (parent) row is deleted:

- (A) The delete operation is rejected
- (B) All matching child rows are automatically deleted as well
- (C) The foreign key value in matching child rows is set to NULL
- (D) Nothing happens to the child rows

**Answer: (B)**

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[VERIFIED PYQ — GATE CSE 2024, Set 2]

**Q57. Relational Algebra - Operations [MCQ · 2 marks]**

The relation schema Person(pid, city) describes the city of residence for every person, uniquely identified by pid. Using only selection, projection, cross product, and rename, what is the minimum number of cross-product operations needed to find the list of cities where at least 3 persons reside?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

**Answer: (B)**

**Solution:** Without aggregate/grouping operators, detecting "at least 3 persons sharing a city" requires comparing three renamed copies of Person pairwise (ensuring their pids are distinct while their cities match). Combining three relations requires exactly 2 cross-product operations: (copy1  $\times$  copy2)  $\times$  copy3, followed by appropriate selections and a projection on city.

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[VERIFIED PYQ — GATE CSE 2006, classic natural-join sizing question]

**Q58. Relational Algebra - Operations [NAT · 2 marks]**

Relations  $r_1(P, Q, R)$  and  $r_2(R, S, T)$  have primary keys P and R respectively.  $r_1$  contains 2000 tuples and  $r_2$  contains 2500 tuples. What is the maximum possible number of tuples in the natural join  $r_1 \bowtie r_2$  (joined on the common attribute R)?

**Answer: 2000**

**Solution:** Since R is the primary key of  $r_2$ , each R-value appears at most once in  $r_2$ . So every tuple of  $r_1$  can match with at most one tuple of  $r_2$ , giving a maximum join size equal to the number of tuples in  $r_1$  (achieved when every R-value of  $r_1$  exists in  $r_2$ ), i.e. 2000.

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[GATE-style]

**Q59. Relational Algebra - Operations [MSQ · 2 marks]**

Which of the following are considered fundamental (primitive) operations of relational algebra, from which other operations (like natural join) can be derived?

- (A) Selection ( $\sigma$ )
- (B) Projection ( $\pi$ )
- (C) Union ( $\cup$ )

(D) Natural Join ( $\bowtie$ )

**Answer: (A), (B), (C)**

**Solution:** (D) Natural join -- along with set intersection and division -- is a derived operation, expressible in terms of the primitive operations (selection, projection, union, set difference, Cartesian product, and rename).

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[VERIFIED PYQ — GATE CSE 2006]

**Q60. Relational Calculus (Tuple & Domain) [MCQ · 2 marks]**

Which of the following relational query languages have the same expressive power? I) Relational algebra. II) Tuple relational calculus restricted to safe expressions. III) Domain relational calculus restricted to safe expressions.

- (A) II and III only
- (B) I and II only
- (C) I and III only
- (D) I, II and III

**Answer: (D)**

**Solution:** This is Codd's theorem -- relational algebra, safe tuple relational calculus, and safe domain relational calculus are all equivalent in expressive power ("relationally complete").

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[VERIFIED PYQ — GATE CSE 2014, Set 3, concept]

**Q61. Relational Calculus (Tuple & Domain) [MCQ · 1 mark]**

Which of the following tuple relational calculus expressions is equivalent to  $\forall t \in r (P(t))$ ?

- (A)  $\neg \exists t \in r (\neg P(t))$
- (B)  $\exists t \in r (\neg P(t))$
- (C)  $\neg \exists t \in r (P(t))$
- (D)  $\exists t \notin r (P(t))$

**Answer: (A)**

**Solution:** "For all tuples  $t$  in  $r$ ,  $P(t)$  holds" is logically equivalent to "there does not exist a tuple  $t$  in  $r$  for which  $P(t)$  is false" -- a standard quantifier-negation identity.

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[New Practice]

**Q62. Relational Calculus (Tuple & Domain) [NAT · 1 mark]**

Relation  $R(A,B,C)$  has 100 tuples. A TRC query  $\{t.A \mid R(t) \wedge t.B = 5\}$  finds 15 matching tuples. How many tuples does the result relation contain (assuming all resulting A-values happen to be distinct)?

**Answer: 15**

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[VERIFIED PYQ — GATE CS 2015, Set 1]

**Q63. SQL Basics [MCQ · 1 mark]**

The SELECT operation in SQL is equivalent to which relational algebra operation?

- (A) The selection operation in relational algebra
- (B) The selection operation in relational algebra, except that SQL's SELECT retains duplicate rows
- (C) The projection operation in relational algebra
- (D) The projection operation in relational algebra, except that SQL's SELECT retains duplicate rows

**Answer: (D)**

**Solution:** SQL's SELECT clause (choosing columns) corresponds to the relational algebra PROJECTION operation ( $\pi$ ), not selection ( $\sigma$ ). However, unlike relational algebra's projection (which removes duplicate tuples under set semantics), SQL's SELECT retains duplicate rows by default (bag/multiset semantics)

unless DISTINCT is explicitly used.

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[VERIFIED PYQ — GATE CSE 2019, concept]

**Q64. SQL Basics [MCQ · 2 marks]**

Given schemas employees(emp-id, first-name, last-name, hire-date, dept-id, salary) and departments(dept-id, dept-name, manager-id, location-id), to display the last names and hire dates of the MOST RECENTLY HIRED employee in each department at location-id 1700, which SQL technique is essential?

- (A) A simple WHERE clause only, with no subquery
- (B) A subquery using MAX(hire-date) combined with GROUP BY dept-id, matched via IN against (dept-id, hire-date) pairs
- (C) ORDER BY hire-date DESC alone, with no grouping
- (D) DISTINCT alone, with no subquery

**Answer: (B)**

**Solution:** Since the maximum hire-date is needed PER department, a grouped subquery (SELECT dept-id, MAX(hire-date) ... GROUP BY dept-id) is required, and the outer query matches against these (dept-id, hire-date) pairs -- the essential technique in this classic GATE query pattern.

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[New Practice]

**Q65. SQL Basics [MSQ · 2 marks]**

Which of the following are valid uses of the WHERE clause?

- (A) Filtering rows based on a numeric comparison, e.g. WHERE age > 18
- (B) Filtering rows based on a string pattern using LIKE
- (C) Combining multiple conditions using AND / OR
- (D) Filtering aggregated group results, e.g. WHERE COUNT(\*) > 5

**Answer: (A), (B), (C)**

**Solution:** (D) is false -- filtering based on an aggregate function's result must be done with HAVING, not WHERE.

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